CYP		Substrats	Inhibiteurs	Inducteurs
CYP1A2		**Théophylline** Certains antiépileptiques Halopéridol Caféine Certains antipsychotiques (**Clozapine**)	Ciprofloxacine Fluvoxamine **Inflammation sévère** (Pneumonie, …) Arrêt du tabac	**Tabac : Goudron**
CYP2C	CYP2C19	**IPP** ***Clopidogrel (pro-drug)*** Antiépileptiques Indométacine Escitalopram	**Antifongiques** (**azoles**) **Cimetidine** **Isoniazid** **IPP** (e.g., **omeprazole**)	**Carbamazepine** **Rifampicin** **Millepertuis** Prednisolone
	CYP2C9	**AINS** **Acénocoumarol (Sintrom)** Antiépileptiques Antidiabétiques oraux (e.g., rosiglitazone) Angiotensin II antagonists	**Sulfonamides** (e.g., sulfamethoxazole) **Sodium valproate** **Metronidazole** **Amiodarone** **Antifongiques** (**azoles**) Chloramphénicol	**Carbamazepine** **Rifampicin** **Millepertuis** **Griseofulvin**
CYP2D6		Beta-bloquants (e.g., **métoprolol**) Antipsychotiques (e.g., **halopéridol**) Certains **antidépresseurs** : SSRI (e.g., fluoxetine, fluvoxamine), SNRI (e.g., venlafaxine), et Tricyclique (e.g., amitriptyline) Antiarythmiques (e.g., flécaïnide) ***Tamoxifène (pro-drug)*** ***Codéine (pro-drug)***	Amiodarone Antipsychotiques and **antidépresseurs** (e.g., **fluoxetine**, paroxetine, duloxetine) Metoclopramide	**Rifampicin** Dexaméthasone
CYP2E1		**OH**	**Disulfiram** Ethanol (aigu)	**Isoniazide** **OH** (chronique) Nicotine
CYP3A4		**Méthadone** **Statines** (sauf pravastatine) Hormones **stéroïdiennes** (e.g., estrogènes, glucocorticoïdes, testostérone) **Contraceptifs oraux** **Carbamazepine** et **autres antiépileptiques** **Haloperidol** **Immunosuppresseurs** **Cyclosporine** **Tacrolimus** **Macrolides** **Benzodiazépines** **Anticoagulants oraux (-xaban)** **AVK** (Acénocoumarol (Sintrom), Phenprocoumon (Marcoumar)) Anti-HIV Anticalciques Itraconazole ***Codéine (pro-drug)***	**Amiodarone** **Macrolides** (érythromycine, clarithromycine) Certains **anti-HIV** (e.g., ritonavir) **Grapefruit juice** **Sodium valproate** **Ritonavir (Paxlovid)** Cimétidine Chloramphénicol **Antifongiques** (**azoles**)	**Carbamazepine** **Rifampicin** **Millepertuis** Certains anti-HIV Griséofulvine Barbituriques (e.g., phénobarbital) Phénytoïne (aussi substrat) Glucocorticoïdes
Pgp		**Indinavir** **Rivaroxaban**	**Ciclosporine**	**Rifampicine**

Interactions Pharmacocinétiques : (modification de l'effet par modification de la concentration plasmatique (ADME))

• Absorption :
- Interaction physico-chimique: ex: quinolones et antiacides ou cations
- Inhibition/ induction de la Pgp
- Inhibition/induction CYP450 intestinaux

• Métabolisme : CYP p450
- Inducteurs*: exemples : rifampicine, carbamazépine (antiépileptiques), millepertuis… → Chercher si manque d'effet
- Inhibiteurs*: exemples : AB macrolides, « azolés », anti VIH, grapefruit → Chercher si effet indésirable ou toxicité

→* sauf *pro-drug* =inverse : exemple : *codéine*, *clopidogrel*

→NB : les métaboliseurs lents ou rapides des CYP450 peuvent avoir fonctionnellement le même effet que inhibiteurs/inducteurs

Interactions Pharmacodynamiques : (modification de l'effet sans modification de la concentration plasmatique →réfléchir en mécanisme d'action)

• 2molécules qui ont un effet net similaire: exemples :
- un anticoagulant +un antiagrégant plaquettaire → saignement
- IEC + spironolactone → hyperkaliémie
- IEC + AINS → insuffisance rénale
- BZD +antipsychotiques → sédation………etc

•2molécules qui ont un effet net inverse

Agents		Indications	Mechanism of action	Adverse effects	Contraindications	Interactions
St. John's Wort		• Major depressive disorder	• Downregulation of β-adrenergic receptors and inhibition of glutamate release [1]	• Gastrointestinal upset • Dizziness, headache, confusion	• Use with caution in bipolar disorder (↑ risk of provoking mania)	• Other serotonergic drugs: ↑ risk of serotonin syndrome • St John's Wort: induces cytochrome P450
Selective serotonin reuptake inhibitors (SSRIs; e.g., fluoxetine, sertraline, citalopram)		• Major depressive disorder (first-line: SSRIs) • Generalized anxiety disorder • Neuropathic pain, stress incontinence, fibromyalgia (SNRIs)	• Inhibition of serotonin reuptake in synaptic cleft → ↑ serotonin levels	• **Sexual dysfunction** • Gastrointestinal upset • Agitation, insomnia, anxiety • ↑ Blood pressure (SNRIs)	• Bleeding disorders • Epilepsy • SNRIs: uncontrolled hypertension	
Serotonin norepinephrine reuptake inhibitors (SNRIs; e.g., venlafaxine, duloxetine)			• Inhibition of serotonin and norepinephrine reuptake in synaptic cleft → ↑ serotonin and norepinephrine levels			
Tricyclic antidepressants (TCAs)	**Secondary amines (e.g., nortriptyline, desipramine)**	• Major depressive disorder • Neuropathic pain • Migraine prophylaxis		• Anticholinergic effects • Cardiotoxicity: prolonged QT interval and widened QRS complex	• Tertiary amines should be avoided in elderly individuals.	
	Tertiary amines (e.g., amitriptyline, imipramine)					
Serotonin antagonist and reuptake inhibitors (SARIs; e.g., trazodone)		• Insomnia • Major depressive disorder (high doses required)	• Inhibition of H_1, α_1, and postsynaptic 5-HT_2 receptors as well as serotonin reuptake → ↑ serotonin levels	• **Priapism** • Sedation	• Avoid in individuals with risk factors for prolonged QT interval. [2]	
Monoamine oxidase inhibitors (MAOIs; e.g., tranylcypromine, phenelzine, selegiline)		• Major depressive disorder (especially atypical depression) • Parkinson disease (selegiline)	• Nonselective inhibition of monoamine oxidase → ↑ levels of epinephrine, norepinephrine, serotonin, and dopamine • Selegiline: selective MAO-B inhibitor → ↑ levels of dopamine	• CNS stimulation • Tyramine-induced hypertensive crisis	• Pheochromocytoma [3]	
Atypical antidepressants	**Mirtazapine**	• Major depressive disorder (mirtazapine: especially individuals who are underweight and/or have insomnia) • Smoking cessation (bupropion, varenicline)	• α_2, H_1, 5-HT_2, and 5-HT_3 receptor antagonists → ↑ serotonin and norepinephrine release and ↑ effect of serotonin on free 5-HT_1 receptors	• Increased appetite, **weight gain**, and serum cholesterol and triglyceride levels • **Sedation**	• Hypersensitivity [4]	
	Bupropion [5]		• Thought to increase dopamine and norepinephrine levels via reuptake inhibition	• **Reduction of seizure threshold** • Insomnia, agitation, tachycardia	• Seizure disorder • Anorexia nervosa, bulimia nervosa	
	Vilazodone		• Inhibition of serotonin reuptake in synaptic cleft → ↑ serotonin levels • 5-HT_{1A} receptor partial agonist	• Sexual dysfunction • Sleep disturbance, abnormal dreams • Gastrointestinal upset	• Concomitant MAOI use [6]	
	Vortioxetine		• Inhibition of 5-HT_3 receptors and serotonin reuptake in synaptic cleft → ↑ serotonin levels • 5-HT_{1A} receptor agonist		• Hypersensitivity	
	Varenicline		• Nicotinic ACh receptor partial agonist	• Sleep disturbances		• Transdermal nicotine: ↑ adverse events
	Buspirone	• Anxiety disorder	• 5-HT_{1A} receptor stimulation	• Headaches, dizziness, nausea, • Seizures resulting from chronic use		• Erythromycin, itraconazole, nefazodone: ↑↑ plasma buspirone concentration [7]